

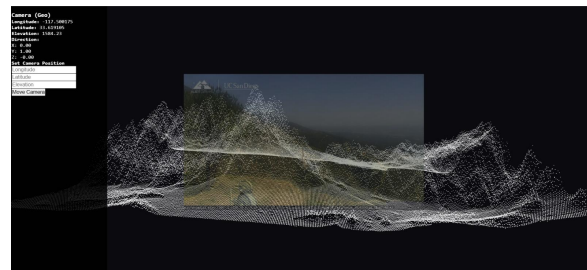
Current status

Data extraction



Extracting images and camera metadata from given video

Frontend



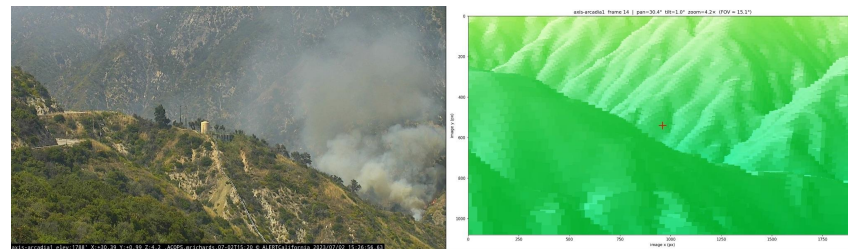
Movable overlay for aligning terrain projection to camera image

Specifications

- Camera: AXIS Q6135-LE PTZ, 1920×1080, 58.3° Horiz. FOV w/o zoom
- Field of view decreases linearly with zooming
- Elevation data: USGS 3DEP via py3dep, 10m resolution, no API required
- Alignment model: 2D similarity transform — translate, zoom, rotate
- Dataset: 22 frames from Chantry Fire, Arcadia CA (~330m from camera)
- Architecture: React (Vite) frontend + FastAPI backend
- Deployed via Helm on Kubernetes

Layout of complete architecture

Backend



Rendering of camera vies from Digital Elevation Map

Questions

- Can we get more data with actual bounding boxes (camera intrinsics)?